

Complex Analysis

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Chapter 1

Cauchy's Theorem

1.1 Goursat's Theorem

Intuition: Recall the Fundamental Thm of Calculus for Contour Integrals implies that: If γ is a closed curve in an open set $\Omega \subseteq \mathbb{C}$ and f is continuous and has a primitive in Ω , then

$$\int_{\gamma} f(z)dz = 0$$

We naturally ask whether the requirements for f can be simplified. We may do so by restricting γ .

Theorem 1.1.1. *Goursat's Theorem*

Let Ω be an open subset of $\mathbb{C}$ and let $T \subseteq \Omega$ be a triangle with interior contained in Ω . If f is holomorphic in Ω , then

$$\int_T f(z)dz = 0$$

Proof. We denote by $T^{(0)}$ the original triangle. Then we can get four smaller triangles by $T_1^{(1)}, T_2^{(1)}, T_3^{(1)}, T_4^{(1)}$ by connecting the midpoints of the three sides of $T^{(0)}$. Then

$$\int_T f(z)dz = \int_{T^{(0)}} f(z)dz = \sum_{j=1}^4 \int_{T_j^{(1)}} f(z)dz$$

Hence, $\exists j \in \{1, 2, 3, 4\}$, s.t.: $|\int_T f(z)dz| \leq 4|\int_{T_j^{(1)}} f(z)dz|$. Denote $T^{(1)} := T_j^{(1)}$ and define $\mathcal{T}^{(1)}$ to be the solid triangle with boundary $T^{(1)}$. Repeat this process, and then we can get a sequence of solid triangles $\mathcal{T}^{(0)} \supseteq \mathcal{T}^{(1)} \supseteq \dots \supseteq \mathcal{T}^{(n)} \supseteq \dots$, such that

$$|\int_T f(z)dz| \leq 4^n |\int_{T^{(n)}} f(z)dz|$$

Note that the diameters of $\{T^{(n)}\}_{n=0}^{\infty}$ converge to 0. Hence, by the Nested Thm, $\{z_0\} = \bigcap_{n=0}^{\infty} \mathcal{T}^{(n)}$ for some $z_0 \in \Omega$. Since f is holomorphic at z_0 , then we can expand f at z_0 into a first-order

approximation:

$$f(z) = f(z_0) + f'(z_0)(z - z_0) + \psi(z)(z - z_0)$$

where $\psi(z)$ denotes the higher-order infinitesimal, i.e.: $\psi(z) \rightarrow 0$ as $z \rightarrow z_0$. Since the constant function $f(z_0)$ and the linear function $f'(z_0)(z - z_0)$ have primitives, then by the Fundamental Thm of Calculus for Contour Integrals, we have:

$$\int_{T^{(n)}} f(z)dz = \int_{T^{(n)}} \psi(z)(z - z_0)dz$$

Let p_n denote the perimeter of $T^{(n)}$ and d_n denote the diameter of $T^{(n)}$. Then $p_n = 2^{-n}p_0$ and $d_n = 2^{-n}d_0$. Therefore,

$$\left| \int_T f(z)dz \right| \leq 4^n \left| \int_{T^{(n)}} f(z)dz \right| = 4^n \left| \int_{T^{(n)}} \psi(z)(z - z_0)dz \right| < p_0 d_0 \epsilon$$

□

Corollary 1.1.2. *of Goursat's Theorem*

Let f be holomorphic in an open set $\Omega \subseteq \mathbb{C}$ that contains a rectangle R and its interior. Then

$$\int_R f(z)dz = 0$$

Proof. After connecting one diagonal of R , we can get two triangles T_1 and T_2 with

$$\int_R f(z)dz = \int_{T_1+T_2} f(z)dz = \int_{T_1} f(z)dz + \int_{T_2} f(z)dz \stackrel{\text{Goursat's}}{=} 0$$

□

1.2 Cauchy Theorem in Discs

Motivation: Goursat's Thm only gives conclusion on triangular path, but we need to calculate on some more general cases. We still need to go back to the coro of the Fundamental Thm of Calculus for Contour Integrals.

Lemma 1.2.1. *Local Existence-of-Primitives Lemma*

A holomorphic function in an open disc has a primitive in that disc.

Proof. Let D be an open disc, WLOG, centered at the origin and let f be holomorphic in D . Note that $\forall z \in D$, we have a path $\gamma_{0,z}$ consisting of two line segments, one horizontal and the other vertical. Define

$$F(z) := \int_{\gamma_{0,z}} f(w)dw$$

Then we claim that F is holomorphic in D and that $F'(z) = f(z)$. $\forall z \in D$, since D is open, then

$\exists r > 0$, s.t.: $B_r(z) \subseteq D$. Hence, $\exists h \in \mathbb{C}$, s.t.: $z + h \in D$. Then we have:

$$F(z + h) - F(z) := \int_{\gamma_{0,z+h}} f(w)dw - \int_{\gamma_{0,z}} f(w)dw \stackrel{\text{Gourset's}}{=} \int_{\eta} f(w)dw \quad (*)$$

where η denotes the line segment from z to $z + h$. By the continuity of f , we can write $f(w) = f(z) + \psi(w)$ where ψ denotes the error term, i.e.: $\psi(w) \rightarrow 0$ as $w \rightarrow z$. Therefore,

$$F(z + h) - F(z) \stackrel{(*)}{=} \int_{\eta} f(w)dw = f(z) \int_{\eta} dw + \int_{\eta} \psi(w)dw = f(z)h + \int_{\eta} \psi(w)dw$$

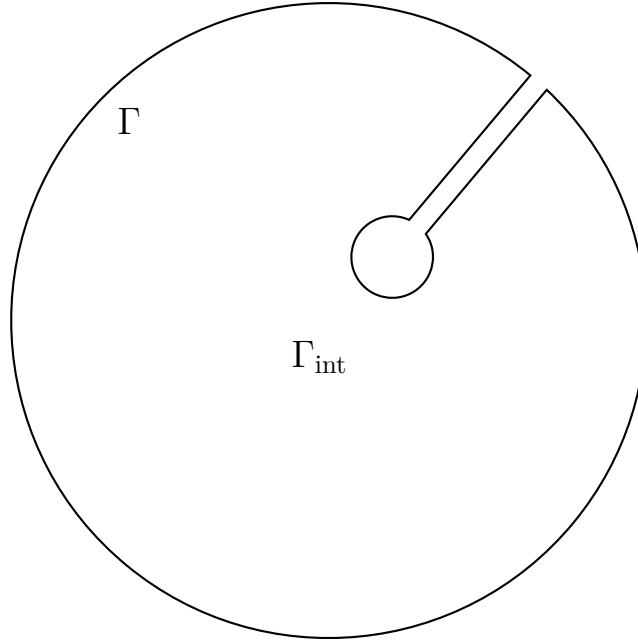
Since $|\int_{\eta} \psi(w)dw| \leq \sup_{w \in \eta} |\psi(w)| \cdot |h|$, then

$$F'(z) = \lim_{h \rightarrow 0} \frac{F(z + h) - F(z)}{h} = f(z)$$

□

Remark: In fact, the proof of Local Existence-of-Primitives Lemma holds not only for discs, but for the region with a good shape which allows every point z in it yielding $\gamma_{0,z}$. We call a *toy contour* any closed curve where the notion of interior is obvious and a construction similar to that in the proof of Local Existence-of-Primitives Lemma is possible in a neighborhood of the curve and its interior.

Then we use an important example of toy contours to prove the Cauchy's Integral Formula later – keyhole contours Γ , consisting of two almost complete circles, one larger and one smaller, connected by a narrow corridor.



Theorem 1.2.2. *Cauchy's Theorem in Disc*

Let f be a holomorphic function in a disc D . Then $\forall$ closed curve γ in D , we have

$$\int_{\gamma} f(z)dz = 0$$

Proof. It's immediately from the Fundamental Thm of Calculus for Contour Integrals and the Local Existence-of-Primitives Lemma. $\square$

Corollary 1.2.3. *of Cauchy's Theorem in Discs*

If f is holomorphic in an open subset of $\mathbb{C}$ containing the circle C and its interior. Then

$$\int_C f(z)dz = 0$$

Proof. Let D be the open disc with boundary C . Then there is a slightly larger disc D' containing D , such that f is holomorphic in D' . We apply Cauchy's Thm in D' and then we're done. $\square$

1.3 Cauchy's Integral Formula

Theorem 1.3.1. *Cauchy's Integral Formula (CIF)*

Let f be holomorphic in an open set that contains the closure of an open disc D and let C denotes the boundary circle of D with the positive orientation. Then $\forall z \in D$,

$$f(z) = \frac{1}{2\pi i} \int_C \frac{f(\zeta)}{\zeta - z} d\zeta$$

Proof. Consider the keyhole contour $\Gamma_{\delta, \epsilon}$ with δ the width of the corridor, ϵ the radius of the smaller circle C_{ϵ} at z and C the larger circle. Consider the function

$$F(\zeta) := \frac{f(\zeta)}{\zeta - z}$$

which is holomorphic away from the point $\zeta = z$. Hence, we have

$$\int_{\Gamma_{\delta, \epsilon}} F(\zeta) d\zeta = 0$$

by Cauchy's Thm in keyholes. Note that for $\delta \rightarrow 0$, the integral directions on the two sides of corridor are opposite and hence cancel each other out, so we have

$$\int_C F(\zeta) d\zeta + \int_{C_{\epsilon}} F(\zeta) d\zeta = 0$$

as $\delta \rightarrow 0$. Consider $F(\zeta) = \frac{f(\zeta) - f(z)}{\zeta - z} + \frac{f(z)}{\zeta - z}$. Since f is holomorphic at z , then $\lim_{\zeta \rightarrow z} \frac{f(\zeta) - f(z)}{\zeta - z} =$

$f'(z)$, so the first term of $F(\zeta)$ is bounded. As for the second term,

$$\int_{C_\epsilon} \frac{f(z)}{\zeta - z} d\zeta = f(z) \int_1^{2\pi} \frac{1}{\epsilon e^{-it}} (-i\epsilon e^{-it} dt) = -2\pi i f(z)$$

Therefore, as $\delta \rightarrow 0, \epsilon \rightarrow 0$, we have:

$$0 = \int_C F(\zeta) d\zeta + \lim_{\epsilon \rightarrow 0} \int_{C_\epsilon} F(\zeta) d\zeta = \int_C \frac{f(\zeta)}{\zeta - z} d\zeta - 2\pi i f(z)$$

□

Corollary 1.3.2. *1 of CIF*

If f is holomorphic in an open set $\Omega \subseteq \mathbb{C}$, then f has infinitely many complex derivatives in Ω . Furthermore, if $C \subseteq \Omega$ is a circle whose interior is also contained in Ω , then $\forall z \in \text{int } C$,

$$f^{(n)}(z) = \frac{n!}{2\pi i} \int_C \frac{f(\zeta)}{(\zeta - z)^{n+1}} d\zeta$$

Proof. It holds immediately by induction with the basic case $n = 0$ being CIF and an easy computation. □

Corollary 1.3.3. *2 of CIF: Cauchy Inequalities*

If f is holomorphic in an open set that contains the closure of a disc D centered at z_0 , of radius R and of boundary C , then

$$|f^{(n)}(z_0)| \leq \frac{n! \cdot \sup_{z \in C} |f(z)|}{R^n}$$

Proof.

$$|f^{(n)}(z_0)| \stackrel{\text{coro.1}}{=} \left| \frac{n!}{2\pi i} \int_C \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta \right| = \frac{n!}{2\pi} \left| \int_0^{2\pi} \frac{f(z_0 + Re^{i\theta})}{(Re^{i\theta})^{n+1}} Re^{i\theta} d\theta \right| \leq \frac{n!}{2\pi} \cdot \frac{\sup_{z \in C} |f(z)|}{R^n} \cdot 2\pi$$

□

Corollary 1.3.4. *3 of CIF: Power Series Expansion Theorem*

Let f be holomorphic in an open set $\Omega \subseteq \mathbb{C}$. If D is a disc centered at z_0 and whose closure is contained in Ω , then f has a power series expansion at z_0

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

$\forall z \in D$, and the coefficients are given by

$$a_n = \frac{f^{(n)}(z_0)}{n!}, \forall n \geq 0$$

Proof. $\forall z \in D$, by CIF, we have:

$$f(z) = \frac{1}{2\pi i} \int_C \frac{f(\zeta)}{\zeta - z} d\zeta$$

where C denotes the boundary of D . If we write

$$\frac{1}{\zeta - z} = \frac{1}{\zeta - z_0 - (z - z_0)} = \frac{1}{\zeta - z_0} \cdot \frac{1}{1 - \frac{z - z_0}{\zeta - z_0}} = \frac{1}{\zeta - z_0} \sum_{n=0}^{\infty} \left(\frac{z - z_0}{\zeta - z_0} \right)^n$$

then we're done. □

Corollary 1.3.5. *of coro.3 of CIF / Power Series Expansion Theorem*

If f is entire, then its power series converge everywhere in $\mathbb{C}$.

Corollary 1.3.6. *4 of CIF: Liouville Theorem*

If f is entire and bounded, then f is constant.

Proof. Let $|f(z)| \leq B, \forall z \in \mathbb{C}$. Then by the Cauchy Inequalities, we get: $\forall z \in \mathbb{C}, \forall R \in \mathbb{R}$,

$$|f'(z)| \leq \frac{B}{R}$$

Take $R \rightarrow \infty$ and then we're done. □

Corollary 1.3.7. *5 of CIF: Fundamental Theorem of Algebra*

1. *(Existence of roots). Every non-constant polynomial $P(z) = a_n z^n + \dots + a_1 z + a_0$ with complex coefficients has a root in $\mathbb{C}$.*
2. *(The number of roots and factorization). Every polynomial $P(z) = a_n z^n + \dots + a_1 z + a_0$ of degree $n \geq 1$ has precisely n roots in $\mathbb{C}$. If these roots are denoted by $w_1, \dots, w_n$, then P can be factored as*

$$P(z) = a_n(z - w_1)(z - w_2) \cdots (z - w_n)$$

Proof. 1. Suppose on the contrary that P has no roots. Assume $a_n \neq 0$.

Claim: $1/P(z)$ is bounded.

Proof. of claim: Whenever $z \neq 0$, we can write

$$\frac{P(z)}{z^n} = a_n + \left(\frac{a_{n-1}}{z} + \dots + \frac{a_0}{z^n} \right)$$

Since each term above in the parentheses goes to 0 as $|z| \rightarrow \infty$, then $\exists R > 0$, s.t.: whenever $|z| > R$,

$$|P(z)| \geq \frac{|a_n|}{2} |z|^n$$

Hence, for $|z| > R$, $|1/P(z)| \leq \frac{2}{|a_n|} \cdot \frac{1}{|z|^n} < \frac{2}{|a_n|} \cdot \frac{1}{R^n}$, so $1/P(z)$ is bounded when $|z| > R$. For $|z| \leq R$, since P is continuous and has no roots in the disc $|z| \leq R$, then P is bounded. □

Then by Liouville Thm, we have: $1/P(z)$ is a constant, contradiction.

2. By 1, P has a root, say w_1 . Write $z = (z - w_1) + w_1$ and then we get

$$P(z) = b_n(z - w_1)^n + \cdots + b_1(z - w_1) + b_0$$

for some $b_n, \dots, b_1, b_0$ with $b_n = a_n$. Since $P(w_1) = 0$, then $b_0 = 0$. Hence,

$$P(z) = (z - w_1)(b_n(z - w_1)^{n-1} + \cdots + b_1) = (z - w_1)Q(z)$$

where Q is a polynomial of degree $n - 1$. Then by induction, we're done. □

1.4 Identity Theorem

Intuition: We've seen, in the Power Series Expansion Thm, that every holomorphic function can be expanded into power series. Then we must think of a question: What are the characteristics of the zeros of it?

Theorem 1.4.1. Identity Theorem

Let f be a holomorphic function in an open connected region Ω vanishes on a sequence of distinct points with a limit point in Ω . Then f is identically 0 in Ω .

Proof. Let $z_0 \in \Omega$ be a limit point for the sequence $\{z_k\}_{k=1}^\infty$, such that $f(z_k) = 0$. We first consider locally a disc $D \subseteq \Omega$ centered at z_0 and consider the power series expansion of f in D :

$$f(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n$$

If f is not identically zero in D , then $\exists$ a smallest integer m , e.t.: $a_m \neq 0$. Then we can write:

$$f(z) = a_m(z - z_0)^m(1 + g(z - z_0))$$

where $g(z - z_0) \rightarrow 0$ as $z \rightarrow z_0$. Take $z = z_k \neq z_0$ for $k > m$. Then $a_m \neq 0$, $(z_k - z_0)^m \neq 0$ and $1 + g(z_k - z_0) \neq 0$, but $f(z_k) = 0$, contradiction.

Consider $U := \text{int } \{z \in \Omega \mid f(z) = 0\} \neq \emptyset$, which is open. On the other hand, by the continuity and the power series expansion of f , U is closed in Ω . Hence, by the connectivity of Ω , the only open and closed subsets are $\emptyset$ and Ω . Therefore, $U = \Omega$. □

Corollary 1.4.2. of Identity Theorem

Let f and g be holomorphic in an open connected region Ω and let $f(z) = g(z)$ for all z in some non-empty open subset of Ω (or more generally for z in some distinct points with limit point in Ω). Then $f = g$ throughout Ω .

Definition 1.4.3. Analytic Continuation

Let f and F be analytic in open Ω respectively Ω' , with $\Omega \subseteq \Omega'$. We say F is an *analytic continuation of f into Ω'* if f and F agree on Ω .

1.5 More Applications

1.5.1 Morera's Theorem

Theorem 1.5.1. *Morera's Theorem*

Let f be a continuous function in the open disc D such that $\forall$ triangle T contained in D ,

$$\int_T f(z)dz = 0$$

Then f is holomorphic.

Proof. Let z_0 be the center of D and $\forall z_1, z_2 \in D$, let γ_{z_1, z_2} denote the line segment from z_1 to z_2 . Define

$$F(z) := \int_{\gamma_{z_0, z}} f(w)dw$$

Fix $z \in D$. $\forall h \in \mathbb{C}$ with $z + h \in D$, we have

$$F(z + h) - F(z) = \int_{\gamma_{z_0, z+h}} f(w)dw - \int_{\gamma_{z_0, z}} f(w)dw \quad (*)$$

Consider a triangle T with vertices $z_0, z, z + h$. By assumption,

$$\begin{aligned} 0 &= \int_T f(w)dw = \int_{\gamma_{z_0, z}} f(w)dw + \int_{\gamma_{z, z+h}} f(w)dw + \int_{\gamma_{z+h, z_0}} f(w)dw \\ &= \int_{\gamma_{z_0, z}} f(w)dw + \int_{\gamma_{z, z+h}} f(w)dw - \int_{\gamma_{z_0, z+h}} f(w)dw \stackrel{(*)}{=} -(F(z + h) - F(z)) + \int_{\gamma_{z, z+h}} f(w)dw \end{aligned}$$

Hence, by def, $F'(z) = f(z)$, so we're done. $\square$

Remark: Morera's Thm gives a converse of Cauchy's Thm.

1.5.2 Sequences of Holomorphic Functions

Intuition: In Real Analysis, the limit of a sequence of differentiable functions may *not* be differentiable. For example, by Weierstrass, every continuous function (may not be differentiable) on $[0, 1]$ can be approximated uniformly by polynomials.

Theorem 1.5.2. *Holomorphic Limit Theorem*

If $\{f_n\}_{n=1}^\infty$ is a sequence of holomorphic functions that converges uniformly to a function f in every compact subset of Ω , then f is holomorphic in Ω .

Proof. Let D be any disc whose closure is contained in Ω and let T be any triangle in D . By Goursat's, $\forall n$, we have

$$\int_T f_n(z) dz = 0$$

Since $f_n \rightarrow f$ uniformly in $\overline{D}$, then f is continuous and

$$\int_T f_n(z) dz \rightarrow \int_T f(z) dz$$

Hence, $\int_T f(z) dz = 0$. Therefore, by Morera's, we're done. $\square$

Theorem 1.5.3. *Uniform Convergence of Derivatives Theorem*

Under the hypothesis of the Holomorphic Limit Thm, the sequence of derivatives $\{f'_n\}_{n=1}^\infty$ converges uniformly to f' on every compact subset of Ω .

Proof. WLOG, assume $\{f_n\}_{n=1}^\infty$ uniformly converges on all of Ω . Define $\Omega_\delta := \{z \in \Omega \mid \overline{D_\delta(z)} \subseteq \Omega\}$.

Claim: For any holomorphic function F in Ω , we have:

$$\sup_{z \in \Omega_\delta} |F'(z)| \leq \frac{1}{\delta} \sup_{\zeta \in \Omega} |F(\zeta)|$$

Proof. of claim: By CIF, $F'(z) = \frac{1}{2\pi i} \int_{C_\delta(z)} \frac{F(\zeta)}{(\zeta-z)^2} d\zeta$, where $C_\delta(z)$ denotes the circle centered at z and of radius δ . Then by the ML Inequality, we get:

$$|F'(z)| \leq \frac{1}{2\pi} \int_{C_\delta(z)} \frac{|F(\zeta)|}{|\zeta-z|^2} d\zeta \stackrel{\text{(ML)}}{\leq} \frac{1}{2\pi} \sup_{\zeta \in \Omega} |F(\zeta)| \frac{1}{\delta^2} 2\pi\delta = \frac{1}{\delta} \sup_{\zeta \in \Omega} |F(\zeta)|$$

$\square$

Let $F = f_n - f$. Then by claim, we have:

$$\sup_{z \in \Omega_\delta} |f'_n(z) - f'(z)| \leq \sup_{\zeta \in \Omega} |f_n(\zeta) - f(\zeta)| \rightarrow 0$$

$\square$

Remark: We may conclude that the Uniform Convergence of Derivatives Thm can be promoted for $f^{(k)}$.

1.5.3 Holomorphic Functions defined in terms of Integrals

Motivation: Consider a function

$$f(z) := \int_a^b F(z, s) ds$$

where F is holomorphic in $\Omega \subseteq \mathbb{C}$ for z and continuous for s . This integral is easy to show that it is well-defined, but whether f is holomorphic is *not* trivial.

Theorem 1.5.4. *Holomorphic Integral Theorem*

Let $F(z, s)$ be defined for $(z, s) \in \Omega \times [0, 1]$ where Ω is an open subset of $\mathbb{C}$. Suppose F satisfies the following properties:

1. $F(z, s)$ is holomorphic in z for each s .
2. F is continuous on $\Omega \times [0, 1]$.

Then the function f , defined on Ω by $z \mapsto \int_0^1 F(z, s)ds$, is holomorphic.

Proof. $\forall n \geq 1$, consider $f_n(z) := (1/n) \sum_{k=1}^n F(z, k/n)$. Then by property 1, f_n is holomorphic in all of Ω . Let D be any open disc whose closure is contained in Ω . Then $\overline{D} \times [0, 1]$ is compact in $\mathbb{C} \times \mathbb{R}$. Hence, by property 2, F is uniformly continuous on $\overline{D} \times [0, 1]$, so $\forall \epsilon > 0, \exists \delta > 0$, s.t.: whenever $|s_1 - s_2| < \delta$,

$$\sup_{z \in D} |F(z, s_1) - F(z, s_2)| < \epsilon$$

Then $\forall n \geq 1/n, \forall z \in D$, we have:

$$|f_n(z) - f(z)| = \left| \sum_{n=1}^k \int_{\frac{k-1}{n}}^{\frac{k}{n}} (F(z, \frac{k}{n}) - F(z, s))ds \right| \leq \sum_{k=1}^n \int_{\frac{k-1}{n}}^{\frac{k}{n}} |F(z, \frac{k}{n}) - F(z, s)|ds < \sum_{k=1}^n \frac{\epsilon}{n} = \epsilon$$

Hence, $f_n \rightarrow f$ uniformly. Therefore, by the Holomorphic Limit Thm, we're done. $\square$

1.5.4 Schwarz Reflection Principle

Motivation: In Real Analysis, the extension of functions is a common and easy method. But in Complex Analysis, the Identity Thm implies that the analytic continuation is difficult to complete.

Convention: Let Ω be an open subset of $\mathbb{C}$.

- Ω is *symmetric* if $z \in \Omega \Leftrightarrow \bar{z} \in \Omega$.
- $\Omega^+ := \{z \in \Omega \mid \text{im } z > 0\}$, $\Omega^- := \{z \in \Omega \mid \text{im } z < 0\}$.
- $I_\Omega := \Omega \cap \mathbb{R}$, which is the common boundary of both Ω^+ and Ω^- .

Lemma 1.5.5. *Symmetry Principle*

Let f^+ and f^- be holomorphic functions in Ω^+ respectively Ω^- , that extend continuously to I_Ω and that

$$f^+(x) = f^-(x), \forall x \in I_\Omega$$

Then f defined on Ω by

$$f(z) := \begin{cases} f^+(z) & , \text{if } z \in \Omega^+ \\ f^+(z) = f^-(z) & , \text{if } z \in I_\Omega \\ f^-(z) & , \text{if } z \in \Omega^- \end{cases}$$

is holomorphic on all of Ω .

Proof. The continuity of f is implied directly by assumption. Then it suffices to show that f is holomorphic on I_Ω : Let D be any disc centered at a point in I_Ω and contained in Ω . For any triangle T in D , there are three cases:

- $T \cap I_\Omega = \emptyset$. By Goursat's, we have

$$\int_T f(z)dz = 0$$

- Either vertex or side of T is on I_Ω . Let T_ϵ denote the triangle translated ϵ into Ω^+ . Then by the continuity of f and the conclusion in the first case, we have: as $\epsilon \rightarrow 0$,

$$0 \xrightarrow{\text{case 1}} \int_{T_\epsilon} f(z)dz \xrightarrow{\text{cont of } f} \int_T f(z)dz$$

Hence, $\int_T f(z)dz = 0$.

- I_Ω intersects T precisely at the two points. Note that we can easily divide T into some smaller triangles which satisfies the second case and applying the conclusion of the second case, we have:

$$\int_T f(z)dz = 0$$

Therefore, by Morera's, we're done. □

Theorem 1.5.6. *Schwarz Reflection Principle*

Let f be a holomorphic function in Ω^+ such that f extends continuously to I_Ω and that f is real-valued on I_Ω . Then there is a function F holomorphic in all of Ω , s.t.: $F = f$ on Ω^+ .

Proof. Define $F(z) := \overline{f(\overline{z})}$, $\forall z \in \Omega^-$, which is obviously holomorphic in Ω^- . Since f is real-valued on I_Ω , then $\overline{f(x)} = f(x)$, $\forall x \in I$. Hence, F extends continuously up to I_Ω . Therefore, by the Symmetry Principle, we're done. □

1.5.5 Runge's Approximation Theorem

Motivation: In Real Analysis, every continuous function on a compact interval, by Weierstrass's Thm, can be approximated uniformly by polynomials. So we ask: What conditions on a compact set $K \subseteq \mathbb{C}$ guarantee that any function holomorphic in an open neighborhood of this set can be approximated by polynomials on K ?

Theorem 1.5.7. *Runge's Approximation Theorem*

Any function holomorphic in an open neighborhood of a compact set K can be approximated uniformly on K by rational functions whose singularities are in K^C . Furthermore, if K^C is connected, then any function holomorphic in an open neighborhood of K can be approximated uniformly on K by polynomials.

Proof.

Lemma 1.5.8. *1: Cauchy Integral Representation Lemma on Compact Sets*

Suppose f is holomorphic in an open set Ω and that $K \subseteq \Omega$ is compact. Then there exist finitely many segment $\gamma_1, \dots, \gamma_N$ in $\Omega \setminus K$, s.t.: $\forall z \in K$,

$$f(z) = \sum_{n=1}^N \frac{1}{2\pi i} \int_{\gamma_n} \frac{f(\zeta)}{\zeta - z} d\zeta$$

Proof. of lemma 1: Take a grid of length $d := \frac{1}{2\sqrt{2}}d(K, \Omega^C)$ and let $\mathcal{Q} := \{Q_1, \dots, Q_M\}$ denote the finite collection of squares in the grid that intersect K . Define $\gamma_1, \dots, \gamma_N$ by the sides of squares in $\mathcal{Q}$ that do not belong to two adjacent squares in $\mathcal{Q}$. Then $\gamma_n \cap K = \emptyset, \forall n$. By the def of d , we have: $\gamma_n \subseteq \Omega, \forall n$. $\forall z \in K$ that is not on the boundary of a square in $\mathcal{Q}$, $\exists j$, s.t.: $z \in Q_j$. Hence, by Cauchy's,

$$\frac{1}{2\pi i} \int_{\partial Q_m} \frac{f(\zeta)}{\zeta - z} d\zeta = \begin{cases} f(z) & , \text{if } m = j \\ 0 & , \text{if } m \neq j \end{cases}$$

Therefore,

$$f(z) = \sum_{m=1}^M \frac{1}{2\pi i} \int_{\partial Q_m} \frac{f(\zeta)}{\zeta - z} d\zeta$$

Since the integral over two adjacent squares cancels, then

$$f(z) = \sum_{n=1}^N \frac{1}{2\pi i} \int_{\gamma_n} \frac{f(\zeta)}{\zeta - z} d\zeta$$

Finally, since $\gamma_n \subseteq K^C, \forall n$, then by the continuity of f , this relation can be promoted for all $z \in K$. $\square$

Lemma 1.5.9. *2: Rational Approximation of Contour Integrals*

$\forall$ line segment $\gamma \subseteq \Omega \setminus K$, $\exists$ a sequence of rational functions with singularities on γ that approximate the integral $\int_{\gamma} f(\zeta)/(\zeta - z)d\zeta$ uniformly on K .

Proof. of lemma 2: Let $\gamma(t)$ be a parameterization for γ . Then

$$\int_{\gamma} \frac{f(\zeta)}{\zeta - z} d\zeta = \int_0^1 \frac{f(\gamma(t))}{\gamma(t) - z} \gamma'(t) dt$$

Since $\gamma \cap K = \emptyset$, then the integrand $F(z, t) := \frac{f(\gamma(t))}{\gamma(t) - z} \gamma'(t)$ is continuous on $K \times [0, 1]$. Since K is compact, then applying the same argument in the proof of the Holomorphic Integral Thm, we see that the Riemann sums of the integral $\int_0^1 F(z, t) dt$ approximate it uniformly on K . Since each of these Riemann sums is a rational function with singularities on γ , then we're done. $\square$

By lemma 1, f can be represented by finitely many contour integrals on γ_n and then each of them, by lemma 2, can be approximated uniformly by rational functions. Hence, the first part of the Runge's Approximation Thm holds. As for the further part, we need one more lemma:

Lemma 1.5.10. *3: Pole Pushing Lemma*

If K^C is connected and $z_0 \notin K$, then the function $1/(z - z_0)$ can be approximated uniformly on K by polynomials.

Proof. of lemma 3: Choose a point z_1 that is outside a large open disc D centered at the origin and containing K . Then

$$\frac{1}{z - z_1} = -\frac{1}{z_1} \cdot \frac{1}{1 - \frac{z}{z_1}} = \sum_{n=1}^{\infty} -\frac{z^n}{z_1^{n+1}}$$

where the series converges uniformly $\forall z \in K \subseteq D$. Hence, the partial sums of this series are polynomials that approximate uniformly $1/(z - z_1)$ on K . Similarly, $\forall k \in \mathbb{N}_+$, $1/(z - z_k)$ can be approximated uniformly on K by polynomials. Now it suffices to prove that $1/(z - z_0)$ can be approximated uniformly on K by polynomials in $1/(z - z_1)$:

Since K^C is connected and hence path-connected, then $\exists$ a path γ from z_0 to z_1 , contained in K^C . Define

$$\rho := \frac{1}{2}d(K, \gamma)$$

which is positive since K and γ are compact. Then choose finitely many points $z_0 = w_0, w_1, \dots, w_l = z_1$, s.t.: $0 \leq j < l$, we have $w_j \in \gamma$ and

$$|w_j - w_{j+1}| < \rho$$

Fix $w \in \gamma$. $\forall w' \in \gamma$ with $|w - w'| < \rho$, note that

$$\frac{1}{z - w} = \frac{1}{z - w'} \cdot \frac{1}{1 - \frac{w - w'}{z - w'}} = \sum_{n=0}^{\infty} \frac{(w - w')^n}{(z - w')^{n+1}}$$

Hence, $\frac{1}{z - w}$ can be approximated uniformly by $\frac{1}{z - w'}$, since $|\frac{w - w'}{z - w'}| < \frac{\rho}{2\rho} = \frac{1}{2} < 1$. Therefore, we can push $\frac{1}{z - z_0} = \frac{1}{z - w_0}$ to gradually $\frac{1}{z - w_1}, \frac{1}{z - w_2}, \dots$, and finally to $\frac{1}{z - w_l} = \frac{1}{z - z_1}$. $\square$

Then by lemma 3, we can push the singularities of rational functions all the way to infinity, thereby transforming the rational functions into polynomials. $\square$

1.6 Generalization for Cauchy's Theorem

Proposition 1.6.1. Homotopy Invariance of Holomorphic Functions Proposition

Let Ω be an open set. If f is holomorphic in Ω , and if γ_0 and γ_1 are two homotopic curves in Ω , then

$$\int_{\gamma_0} f(z)dz = \int_{\gamma_1} f(z)dz$$

Proof. Let $\gamma_0(t)$ and $\gamma_1(t)$ be the parameterizations of γ_0 respectively γ_1 defined on $[a, b]$ with $\gamma_0(a) = \gamma_1(a) = \alpha$ and $\gamma_0(b) = \gamma_1(b) = \beta$, and let $\gamma : [0, 1] \times [a, b] \rightarrow \mathbb{C}$ be the homotopy from γ_0 to γ_1 . Since $[0, 1] \times [a, b]$ is compact and γ is continuous, then $K := \text{im } \gamma$ is compact and γ is

continuous, so $\exists \epsilon > 0$, s.t.: every disc of radius 3ϵ centered at a point in K is contained in Ω , and $\exists \delta > 0$, s.t.: whenever $|s_1 - s_2| < \delta$,

$$\sup_{t \in [a, b]} |\gamma(s_1, t) - \gamma(s_2, t)| < \epsilon$$

Fix $s_1, s_2 \in [0, 1]$ with $|s_1 - s_2| < \delta$ and then choose discs $\{D_0, \dots, D_n\}$ of radius 2ϵ and points $\{\alpha = z_0, \dots, z_{n+1} = \beta\}$ on $\gamma_{s_1} := \gamma(\{s_1\} \times [a, b])$ and $\{\alpha = w_0, \dots, w_{n+1} = \beta\}$ on $\gamma_{s_2} := \gamma(\{s_2\} \times [a, b])$ such that γ_{s_1} and γ_{s_2} are covered by $\{D_0, \dots, D_n\}$ and that $\forall i = 0, \dots, n$, the points $z_i, z_{i+1}, w_i, w_{i+1}$ are in D_i .

By the Local Existence-of-Primitives Lemma, on each disc D_i , f admits a primitive, denoted as F_i . Then on $D_i \cap D_{i+1}$, since F_i and F_{i+1} are two primitives of the same functions, then $F_{i+1} - F_i$ is a constant function. Hence,

$$F_{i+1}(z_{i+1}) - F_i(z_i) = F_{i+1}(w_{i+1}) - F_i(w_i) \Leftrightarrow F_{i+1}(z_{i+1}) - F_{i+1}(w_{i+1}) = F_i(z_i) - F_i(w_i) \quad (*)$$

Therefore,

$$\begin{aligned} \int_{\gamma_{s_1}} f - \int_{\gamma_{s_2}} f &= \sum_{i=0}^n (F_i(z_{i+1}) - F_i(z_i)) - \sum_{i=0}^n (F_i(w_{i+1}) - F_i(w_i)) \\ &= \sum_{i=0}^n ((F_i(z_{i+1}) - F_i(z_i)) - (F_i(w_{i+1}) - F_i(w_i))) \\ &\stackrel{(*)}{=} F_n(z_{n+1}) - F_n(w_{n+1}) - (F_0(z_0) - F_0(w_0)) \\ &= F_n(\alpha) - F_n(\alpha) - (F_0(\beta) - F_0(\beta)) = 0 \end{aligned}$$

Then by subdividing $[0, 1]$ into subintervals $[s_i, s_{i+1}]$ of length less than δ , we can go from γ_0 to γ_1 by finitely many times applying $\int_{\gamma_{s_i}} f = \int_{\gamma_{s_{i+1}}} f$. $\square$

Lemma 1.6.2. Existence of Primitives in Simply Connected Domain Lemma

Any holomorphic function in a simply connected domain has a primitive.

Proof. Let f be holomorphic in a simply connected domain Ω . Fix $z_0 \in \Omega$. $\forall z \in \Omega$, let γ be any curve in Ω from z_0 to z . By the Homotopy Invariance of Holomorphic Functions Prop, we can well-definedly define

$$F(z) := \int_{\gamma} f(w)dw$$

i.e.: the prop guarantees that this def is independent of the curve γ . If we denote by $\gamma_{z, z'}$ the line segment from z to z' , then we can write

$$F(z+h) - F(z) = \int_{\gamma_{z, z+h}} f(w)dw$$

Applying the same argument in the proof of the Local Existence-of-Primitives Lemma, then we're

done. □

Theorem 1.6.3. *Cauchy's Theorem in Simply Connected Domains*

If f is holomorphic in a simply connected domain Ω , then $\forall$ closed curve γ in Ω ,

$$\int_{\gamma} f(z)dz = 0$$

Proof. It's immediately from the Fundamental Thm of Calculus for Contour Integrals and the Existence of Primitives in Simply Connected Domains Lemma. □

Corollary 1.6.4. *of Cauchy's Theorem in Simply Connected Domains*

The punctured plane $\mathbb{C} \setminus \{0\}$ is not simply connected.

Proof. It's immediately from the Cauchy's Theorem in Simply Connected Domains and the fact that the integral of $1/z$ over the unit circle is $2\pi i$ but not 0. □

Chapter 2

Singularities

2.1 Zeros and Poles

Definition 2.1.1. Isolated Singularities

An *isolated singularity* (or *point singularity*) of a function f is a complex number z_0 such that f is defined in an open neighborhood z_0 except for the point z_0 itself.

Motivation: The singularities of holomorphic functions usually appear when the denominator in a fraction equals to zero, so to fully master singularities, we need to study the zeros of functions first.

Definition 2.1.2. Zeros

A complex number z_0 is a *zero* for the holomorphic function f if $f(z_0) = 0$.

Remark: The identity Thm implies that the zeros of a non-trivial holomorphic function are isolated.

Theorem 2.1.3. Local Factorization Theorem for Zeros

Let f be holomorphic in a connected open subset $\Omega \subseteq \mathbb{C}$. Suppose f has a zero at a point $z_0 \in \Omega$ and doesn't vanish identically in Ω . Then there is an open neighborhood $U \subseteq \Omega$ of z_0 , a non-vanishing holomorphic function g on U , and a unique positive integer n , s.t.: $\forall z \in U$,

$$f(z) = (z - z_0)^n g(z)$$

Proof. In a small disc centered at z_0 , f has a power series expansion $f(z) = \sum_{k=0}^{\infty} a_k(z - z_0)^k$. Since f is not identically zero, then there is a smallest positive integer n , s.t.: $a_n \neq 0$. Then we write

$$f(z) = (z - z_0)^n (a_n + a_{n+1}(z - z_0) + \cdots)$$

Define $g(z) := \sum_{k=0}^{\infty} a_{n+k}(z - z_0)^k$, which is holomorphic and is non-vanishing $\forall z$ closed to z_0 (since $a_n \neq 0$). To see the uniqueness of n , we suppose that $f(z) = (z - z_0)^n g(z) = (z - z_0)^m h(z)$

for some $m \in \mathbb{N}_+$ and some $h(z_0) \neq 0$. Suppose on the contrary that, WLOG, $m > n$. Then $g(z) = (z - z_0)^{m-n}h(z)$, where $z \rightarrow z_0$ yields $g(z_0) = 0$, contradiction. Hence, $m = n$. $\square$

Definition 2.1.4. Order of Zeros

The unique $n \in \mathbb{N}_+$ in the Local Factorization Theorem for Zeros is called the *order* (or *multiplicity*) of the zero z_0 for f . Furthermore, if $n = 1$, then z_0 is said to be *simple*.

Definition 2.1.5. Poles

- A *deleted neighborhood* of z_0 is defined to be an open disc

$$\{z \mid 0 < |z - z_0| < r\}$$

for some $r > 0$.

- A function f defined in a deleted neighborhood of z_0 has a *pole* at z_0 if the function $1/f$, defined to be zero at z_0 , is holomorphic in a full neighborhood of z_0 .

Theorem 2.1.6. Local Representations of Poles Theorem

If f has a pole at $z_0 \in \Omega$, then in an open neighborhood of z_0 , there exist a non-vanishing holomorphic function h and a unique positive integer n , s.t.:

$$f(z) = (z - z_0)^{-n}h(z)$$

Proof. It's immediately from the def of poles and the Local Factorization Thm for Zeros. $\square$

Definition 2.1.7. Order of Poles

The unique n in the Local Representations of Poles Thm is called the *order* (or *multiplicity*) of the pole z_0 for f . Furthermore, if $n = 1$, then z_0 is said to be *simple*.

Theorem 2.1.8. Principal Part Expansions at Poles Theorem

If f has a pole of order n at z_0 , then

$$f(z) = \frac{a_{-n}}{(z - z_0)^n} + \frac{a_{-n+1}}{(z - z_0)^{n-1}} + \cdots + \frac{a_{-1}}{z - z_0} + G(z)$$

where G is a holomorphic function in an open neighborhood of z_0 .

Proof. It's immediately from the Local Representations of Poles Thm by expanding h . $\square$

Definition 2.1.9. Principal Part

The sum

$$\frac{a_{-n}}{(z - z_0)^n} + \frac{a_{-n+1}}{(z - z_0)^{n-1}} + \cdots + \frac{a_{-1}}{z - z_0}$$

in the Principal Part Expansions at Poles Thm is called the *principal part* of f at the pole z_0 .

Definition 2.1.10. Residue

The coefficient a_{-1} in the Principal Part Expansions at Poles Thm is the *residue* of f at the pole, denoted as $\text{res}_{z_0} f := a_{-1}$.

Remark:

- $\forall k = 2, \dots, n$, the term $\frac{a_{-k}}{(z-z_0)^k}$ in the principal part has primitive in a deleted neighborhood of z_0 :

$$\frac{-a_{-k}}{(k-1)(z-z_0)^{k-1}}$$

where integral on any circle C centered at the origin is zero and which is a single-valued function. But for the residue a_{-1} of f , its primitive is $\log(z-z_0)$, which is multi-valued. This is why we are discussing the situation of a_{-1} separately.

- If f has a simple pole at z_0 , then it's clear that

$$\text{res}_{z_0} f = \lim_{z \rightarrow z_0} (z - z_0) f(z)$$

But for the pole of higher order, the form of evaluation of $\text{res}_{z_0} f$ is not trivial:

Theorem 2.1.11. *Residue Computation Theorem*

If f has a pole of order n at z_0 , then

$$\text{res}_{z_0} f = \lim_{z \rightarrow z_0} \frac{1}{(n-1)!} \left(\frac{d}{dz} \right)^{n-1} (z - z_0)^n f(z)$$

Proof. It's immediately from a consequence of the Principal Part Expansions at Poles Thm:

$$(z - z_0)^n f(z) = a_{-n} + a_{-n+1}(z - z_0) + \dots + a_{-1}(z - z_0)^{n-1} + G(z)(z - z_0)^n$$

□

2.2 Residue Formula

Theorem 2.2.1. *Cauchy's Residue Theorem*

Suppose f is holomorphic in an open set containing a circle C and its interior, except for a pole at z_0 inside C . Then

$$\int_C f(z) dz = 2\pi i \text{res}_{z_0} f$$

Proof. Consider a keyhole contour whose small circle C_ϵ is centered at z_0 and of radius ϵ . Let the width of the corridor go to zero and then we have:

$$\int_C f(z) dz = \int_{C_\epsilon} f(z) dz$$

Note that by applying CIF to the constant function a_{-1} , we get:

$$\frac{1}{2\pi i} \int_{C_\epsilon} \frac{a_{-1}}{z - z_0} dz = a_{-1} = \text{res}_{z_0} f$$

that $\forall k > 1$, by applying cor.1 of CIF to the constant function a_{-k} , we get:

$$\frac{1}{2\pi i} \int_{C_\epsilon} \frac{a_{-k}}{(z - z_0)^k} dk = 0$$

and that by Cauchy's Thm, we have: $\int_{C_\epsilon} G(z) dz = 0$, since G is holomorphic. Hence, applying $\int_{C_\epsilon} * dz$ to $f(z)$ with

$$f(z) = \frac{a_{-n}}{(z - z_0)^n} + \frac{a_{-n+1}}{(z - z_0)^{n-1}} + \cdots + \frac{a_{-1}}{z - z_0} + G(z)$$

we're done. □

Corollary 2.2.2. *1 of Cauchy's Residue Theorem*

Suppose f is holomorphic in an open set containing a circle C and its interior, except for poles at the points $z_1, \dots, z_N$ inside C . Then

$$\int_C f(z) dz = 2\pi i \sum_{k=1}^N \text{res}_{z_k} f$$

Proof. Similar as the proof of Cauchy's Residue Thm, except for considering a multiple keyhole contour to pass by each pole. □

Corollary 2.2.3. *2 of Cauchy's Residue Theorem: Residue Formula*

Suppose f is holomorphic in an open set containing a toy contour γ and its interior, except for poles at the points $z_1, \dots, z_N$ inside γ . Then

$$\int_\gamma f(z) dz = 2\pi i \sum_{k=1}^N \text{res}_{z_k} f$$

2.3 Other types of Singularities

Intuition: We've learned something about the poles, so now we need to pay attention to some other types of isolated singularities.

2.3.1 Removable Singularities

Motivation: Consider the function $\frac{(z-1)(z-2)}{z-2}$, which has a singularity at $z = 2$, but can be extended to be $z - 1$.

Definition 2.3.1. Removable Singularities

Let f be a function holomorphic in an open set $\Omega \subseteq \mathbb{C}$, except for the point $z_0 \in \Omega$. We say that

z_0 is a *removable singularity* if we can define f at z_0 in such a way that f becomes holomorphic in all of Ω .

Theorem 2.3.2. *Riemann's Theorem on Removable Singularities*

Suppose f is holomorphic in an open set Ω , except for a point $z_0 \in \Omega$. If f is bounded on $\Omega \setminus \{z_0\}$, then z_0 is a removable singularity.

Proof. Consider an open disc D centered at z_0 whose closure is contained in Ω , let C denote the boundary of D , and fix $z \in D$ with $z \neq z_0$. Consider a multiple keyhole contour with two small circles where z_0 and z are the centers of two circles C_0 respectively C_1 both of radius ϵ . Let the sides of the two corridors go close to each other and finally overlap. Then we get:

$$\int_C \frac{f(\zeta)}{\zeta - z} d\zeta + \int_{C_0} \frac{f(\zeta)}{\zeta - z} d\zeta + \int_{C_1} \frac{f(\zeta)}{\zeta - z} d\zeta = 0$$

As in the proof of CIF, we have:

$$\int_{C_1} \frac{f(\zeta)}{\zeta - z} d\zeta = -2\pi i f(z)$$

Take a very small ϵ such that ζ on C_0 stays from z . Then by the assumption that f is bounded, i.e.: $|f| \leq M$ for some $M > 0$, we have:

$$\left| \int_{C_0} \frac{f(\zeta)}{\zeta - z} d\zeta \right| \stackrel{\text{(ML)}}{\leq} \frac{M}{\sup_{\zeta \in C_0} |\zeta - z|} 2\pi\epsilon \xrightarrow{\epsilon \rightarrow 0} 0$$

Therefore, we can conclude that

$$f(z) = \frac{1}{2\pi i} \int_C \frac{f(\zeta)}{\zeta - z} d\zeta$$

Then by the corollary of Identity Thm, we're done. $\square$

Corollary 2.3.3. *of Riemann's Theorem on Removable Singularities: Equivalent Characterization of Poles Proposition*

Suppose f has an isolated singularity at the point z_0 . Then z_0 is a pole of f iff $|f(z)| \rightarrow \infty$ as $z \rightarrow z_0$.

Proof. $(\Rightarrow)$ holds immediately by the def of poles. Conversely, by assumption, $1/f(z)$ is bounded near z_0 . Hence, $1/f$ has a removable singularity at z_0 and in fact vanish at z_0 . Therefore, by the def of removable singularities, we can extend $1/f$ to be zero at z_0 , so f has a pole at z . $\square$

2.3.2 Essential Singularities

Definition 2.3.4. Essential Singularities

Any singularity that is not removable or a pole is defined to be an *essential singularity*.

Example: $e^{1/z}$ has an essential singularity at $z = 0$: Near $z = 0$, $e^{1/z}$ oscillates and grows faster than any power, and a complete understanding of its behavior is not easy.

Theorem 2.3.5. *Casorati-Weierstrass Theorem*

Suppose f is holomorphic in the punctured disc $D_r(z_0) \setminus \{z_0\}$ and has an essential singularity at z_0 . Then the image of $D_r(z_0) \setminus \{z_0\}$ under f is dense in the complex plane.

Proof. Suppose on the contrary that $\exists w \in \mathbb{C}, \exists \delta > 0$, s.t.: $\forall z \in D_r(z_0) \setminus \{z_0\}, |f(z) - w| > \delta$. Consider the function

$$g(z) := \frac{1}{f(z) - w}$$

with $|g(z)| < \frac{1}{\delta}$ and hence bounded. Then by the Riemann's Thm on Removable Singularities, $g(z)$ has a removable singularity at z_0 :

- If $g(z_0) \neq 0$, then $f(z) = w + \frac{1}{g(z)}$ is holomorphic at z_0 , contradiction.
- If $g(z_0) = 0$, then $f(z) - w = \frac{1}{g(z)}$ has a pole at z_0 , contradiction.

□

2.4 Meromorphic Functions

Definition 2.4.1. Meromorphic Functions

A function f on an open set Ω is *meromorphic* if there exists a sequence of points $\{z_0, z_1, \dots\}$ that has no limit points in Ω satisfying the followings:

- f is holomorphic in $\Omega \setminus \{z_0, z_1, \dots\}$.
- f has poles at the points $\{z_0, z_1, \dots\}$.

Remark (*Meromorphic Functions in the Extended Complex Plane*). If a function f is holomorphic for all large values of z , then the function $F(z) := f(1/z)$ is holomorphic in a deleted neighborhood of the origin.

- We then say that f has a *pole at ∞* if F has a pole at the origin. Similarly, we can speak of f having an *essential singularity at ∞* or a *removable singularity* (hence *holomorphic*) *at ∞* in terms of the corresponding behavior of F at 0.
- A meromorphic function in $\mathbb{C}$, that is either holomorphic at ∞ or has a pole at ∞ , is said to be a meromorphic function in the extended complex plane $\widehat{\mathbb{C}} := \mathbb{C} \cup \{\infty\}$.

Theorem 2.4.2. *Characterization of Meromorphic Functions in the Extended Complex Plane Theorem*

The meromorphic functions in $\widehat{\mathbb{C}}$ are the rational function.

Proof. Let f be meromorphic in $\widehat{\mathbb{C}}$. Then f has finitely many poles $z_1, \dots, z_n$ in $\widehat{\mathbb{C}}$, since otherwise the sequence of infinitely many points in $\widehat{\mathbb{C}}$ must converge to somewhere in $\widehat{\mathbb{C}}$, which is a contradiction to the def of meromorphic functions. $\forall k = 1, \dots, n$, near z_k , let $f_k(z)$ denote the principal part of $f(z)$, and at ∞ , let $\tilde{f}_\infty(z)$ denote the principal part of $f(1/z)$ at 0 and then set

$f_\infty(z) := \tilde{f}_\infty(1/z)$. Consider a function

$$H(z) := f(z) - f_\infty(z) - \sum_{k=1}^n f_k(z)$$

which is entire and bounded in all of $\hat{\mathbb{C}}$. By Liouville's, $H(z) \equiv M$ is a constant function. Hence,

$$f(z) + f_\infty(z) + \sum_{k=1}^n f_k(z) + M$$

Note that every principal part f_k is polynomial in $1/(z - z_k)$ and that $\tilde{f}_\infty$ is a polynomial in $1/z$ which implies that f_∞ is a polynomial in z . Therefore, f is a rational function. $\square$

2.5 Argument Principle

2.5.1 Argument Principle

Intuition: Recall that the function $\log f(z) := \log |f(z)| + i \arg f(z)$ is multi-valued, but the derivative is $f'(z)/f(z)$, which is single-valued. So we are interested in studying f'/f .

Theorem 2.5.1. *Argument Principle*

Suppose f is meromorphic in an open set containing a circle C and its interior. If f has no poles and never vanishes on C , then

$$\frac{1}{2\pi i} \int_C \frac{f'(z)}{f(z)} dz = \#(\text{zeros of } f \text{ inside } C) - \#(\text{poles of } f \text{ inside } C)$$

where the zeros and poles are counted with their multiplicities.

Proof. • If f is holomorphic and has a zero of order n at z_0 , we write

$$f(z) = (z - z_0)^n g(z)$$

where g is holomorphic and non-vanishing in a neighborhood of z_0 . Hence,

$$\frac{f'(z)}{f(z)} = \frac{n}{z - z_0} + \frac{g'(z)}{g(z)}$$

has a simple pole with residue n at z_0 .

- If f is holomorphic and has a pole of order n at z_0 , then we write $f(z) = (z - z_0)^{-n} h(z)$ and then we have

$$\frac{f'(z)}{f(z)} = \frac{-n}{z - z_0} + \frac{h'(z)}{h(z)}$$

having a simple pole with residue $-n$ at z_0 .

Therefore, if f is meromorphic, then f'/f have simple poles at the zeros and poles of f , with residue = the order of the zero of f or the negative of the order of the pole of f . $\square$

Remark: This thm holds also for toy contours.

2.5.2 Applications of Argument Principle

Rouche's Theorem

Theorem 2.5.2. *Rouche's Theorem*

Suppose f and g are holomorphic in an open set containing a circle C and its interior. If $\forall z \in C$,

$$|f(z)| > |g(z)|$$

then f and $f + g$ have the same number of zeros inside C .

Proof. For $t \in [0, 1]$, define $f_t(z) := f(z) + tg(z)$. Then $f_0 = f$ and $f_1 = f + g$. Let n_t denote the number of zeros of f_t inside C counted with multiplicities. $\forall z \in C$, since $|f(z)| > |g(z)|$, then

$$|f_t(z)| \geq |f(z)| - t|g(z)| \geq |f(z)| - |g(z)| > 0$$

Hence, f_t has no zeros on C , so by the Argument Principle, we have

$$n_t = \frac{1}{2\pi i} \int_C \frac{f'_t(z)}{f_t(z)} dz$$

Therefore, n_t is continuous for $t \in [0, 1]$. By the def of n_t , we then see that it must be a constant. $\square$

Open Mapping Theorem

Theorem 2.5.3. *Open Mapping Theorem*

If f is holomorphic and non-constant in an open set $\Omega \subseteq \mathbb{C}$, then f is open.

Proof. $\forall w_0 \in \text{im } f, \exists z_0 \in \Omega$, s.t.: $f(z_0) = w_0$. Define

$$g(z) := f(z) - w, F(z) := f(z) - w_0 \text{ and } G(z) := w_0 - w$$

Then $g = F + G$. Since Ω is open and f is not constant, then $\exists \delta > 0$, s.t.: the closure of the disc $D : |z - z_0| < \delta$ is contained in Ω and that $f(z) \neq w_0$ on the boundary C of D . Then we take $\epsilon > 0$ to be such that $|f(z) - w_0| \geq \epsilon$ on C . Consider $B_\epsilon(w_0)$. $\forall w \in B_\epsilon(w_0)$, $|G(z)| < \epsilon \leq |F(z)|$ on C , so can apply Rouché's Thm to conclude that F has the same number of zeros of $g = F + G$ inside C , say g has at least one zero inside C , i.e.: $\forall w \in B_\epsilon(w_0), \exists z \in D \subseteq \Omega$, s.t.:

$$f(z) = w$$

which implies that $B_\epsilon(w) \subseteq \text{im } f$ and hence gives the openness of $\text{im } f$. $\square$

Maximum Modulus Principle

Definition 2.5.4. Maximum of Holomorphic Function

The *maximum* of a holomorphic function f in an open set Ω is defined to be the maximum of its absolute value $|f|$ in Ω .

Theorem 2.5.5. Maximum Modulus Principle

If f is a non-constant holomorphic function in an open set $\Omega \subseteq \mathbb{C}$, then f cannot attain a maximum in Ω .

Proof. Suppose on the contrary that f attain a maximum at $z_0 \in \Omega$. By the Open Mapping Thm, f is open. Hence, $f(D)$ is open and is containing $f(z_0)$, where $D \subseteq \Omega$ is an arbitrary open disc containing z_0 . Since f is non-constant, then we can find some D , s.t.: $\exists z \in D$ with $|f(z)| > |f(z_0)|$, contradiction. $\square$

Corollary 2.5.6. of Maximal Modulus Principle

Suppose that Ω is an open set with compact closure. If f is holomorphic on Ω and continuous on $\overline{\Omega}$, then

$$\sup_{z \in \Omega} |f(z)| \leq \sup_{z \in \overline{\Omega} \setminus \Omega} |f(z)|$$